```c
#include <stdio.h>
#include <math.h>

int main() {
  float a, b, c;

  // Taking input from user
  printf("Enter the three sides of the triangle: ");
  scanf("%f %f %f", &a, &b, &c);

  // Check for validity using triangle inequality theorem
  if (a + b > c && a + c > b && b + c > a) {
    printf("Triangle is valid.\n");

    // Check for Equilateral
    if (a == b && b == c)
      printf("It is an Equilateral Triangle.\n");

    // Check for Isosceles
    else if (a == b || b == c || a == c)
      printf("It is an Isosceles Triangle.\n");

    // Check for Right-angled triangle
    if (fabs(a*a + b*b - c*c) < 0.0001 ||
      fabs(a*a + c*c - b*b) < 0.0001 ||
      fabs(b*b + c*c - a*a) < 0.0001)
      printf("It is a Right-Angled Triangle.\n");
```

```
PS C:\Users\ACER\Desktop\itscbe> gcc 3.1.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
Enter the three sides of the triangle: 45
4
5
Triangle is NOT valid.
PS C:\Users\ACER\Desktop\itscbe>
```

Experiment 3.1 : Conditional statements

1. WAP to check if the triangle is valid or not. If the validity is established do check if the triangle is isosceles, equilateral, right angle, or scalene.
Take sides of the triangle as input from the user.

→
```
#include<stdio.h>

int main() {
 int a,b,c;

printf("Enter 3 sides of a triangle: ");
scanf("%d,%d,%d", &a, &b, &c);

if((a+b>c) && (a+c>b) && (b+c>a)) {
   printf("Valid Triangle \n");

  if(a==b && b==c) {
     printf("Equilateral Triangle \n");
  }
  else if(a==b || b==c || a==c) {
     printf("Isosceles Triangle \n");
  }
  else {
```

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```c
printf("Scalene Triangle\n");
}
if ((a*a + b*b==c*c) ||
   (a*a +c* c== b*b) ||
   (b*b+ c* c ==a*a)) {
   printf("Right-angle Triangle \n");
  }
} else {
    printf("Not a valid Triangle \n");
}
return0;
}
```

2. WAP to compute the BMI Index of the person and print t BMI values as per the following ranges. You can use followin formula to compute BMI= weight (kgs) / Height (Mts) * Height (Mts)

$$BMI = \frac{weight\,(kgs)}{Height\,(Mts) * Height\,(Mts)}$$

→

```c
#include<stdio.h>

int main() {
    float weight, height, bmi;

printf("Enter weight in kg : ");
scanf("%f, &weight);
```

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```c
int main() {
    float l1, b1, l2, b2, l3, b3;
    float p1, p2, p3, max;

    // Input length and breadth of three rectangles
    printf("Enter length and breadth of Rectangle 1: ");
    scanf("%f %f", &l1, &b1);

    printf("Enter length and breadth of Rectangle 2: ");
    scanf("%f %f", &l2, &b2);

    printf("Enter length and breadth of Rectangle 3: ");
    scanf("%f %f", &l3, &b3);

    // Calculate perimeters
    p1 = 2 * (l1 + b1);
    p2 = 2 * (l2 + b2);
    p3 = 2 * (l3 + b3);

    // Find maximum using ternary operator
    max = (p1 > p2) ? ((p1 > p3) ? p1 : p3) : ((p2 > p3) ? p2 : p3);

    printf("\nPerimeter of Rectangle 1: %.2f\n", p1);
    printf("Perimeter of Rectangle 2: %.2f\n", p2);
    printf("Perimeter of Rectangle 3: %.2f\n", p3);

    // Determine which rectangle has the highest perimeter
    (max == p1) ? printf("\nRectangle 1 has the highest perimeter.\n") :
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
43
Enter length and breadth of Rectangle 3: 32
22

Perimeter of Rectangle 1: 154.00
Perimeter of Rectangle 2: 132.00
Perimeter of Rectangle 3: 108.00

Rectangle 1 has the highest perimeter.
PS C:\Users\ACER\Desktop\itscbe>
```

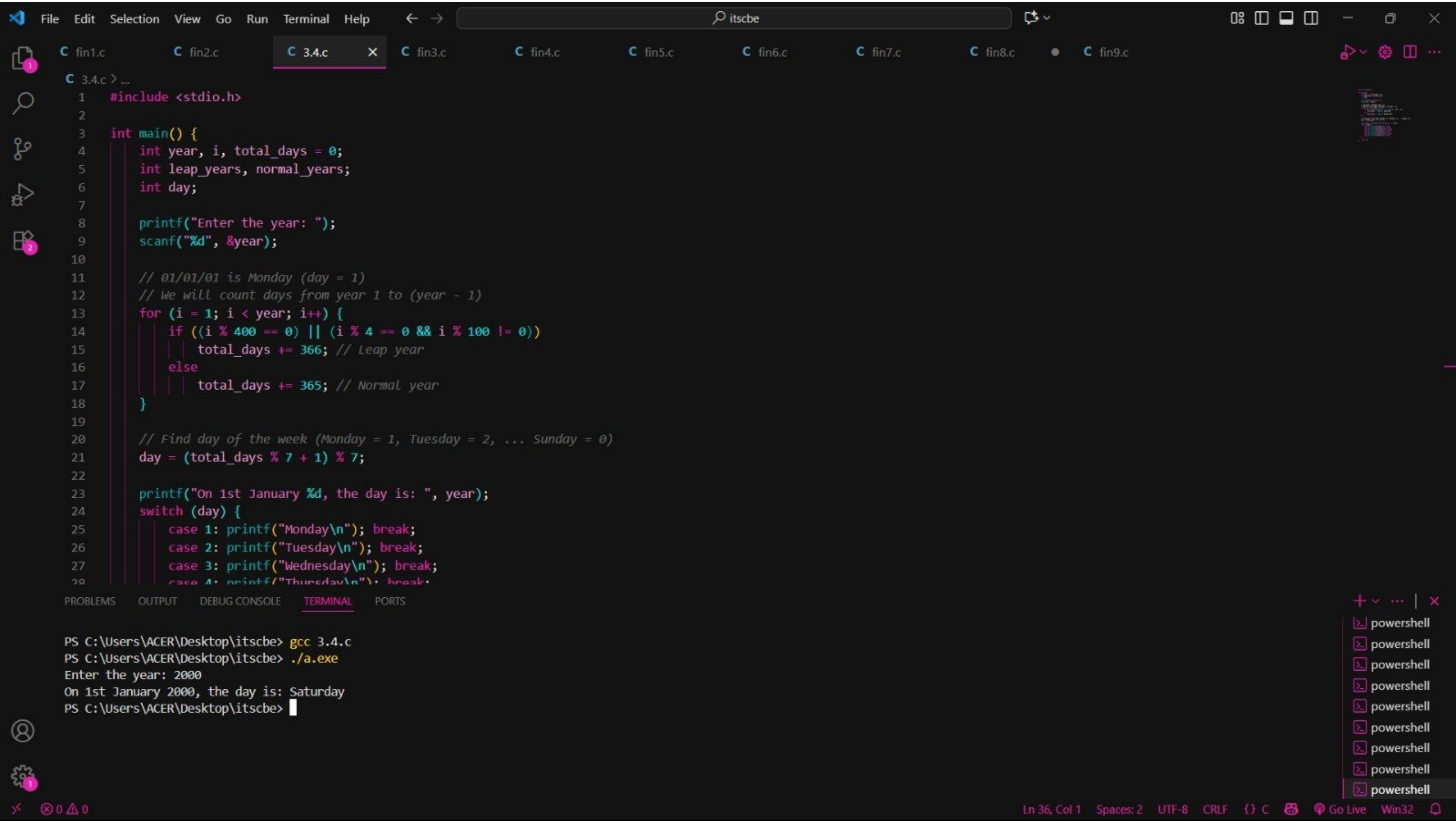
```
case 0 : printf("Sunday \n"); break;
case 1 :printf("Monday \n"); break;
case 2 : printf("Tuesday \n"); break;
case 3: printf("Wednesday \n"); break;
case 4: printf("Thursday \n"); break;
case 5 : printf("Friday \n"); break;
case 6: printf ("Saturday \n"); break;

}
  return 0;
}
```

5. WAP using ternary operator, the user should input the length and breadth of a rectangle, one has to find out which rectangle has the highest perimeter. The minimum number of rectangles should be three.

→

```
#include<stdio.h>

int main() {
    int l1, b1, l2, b2, l3, b3;
    int p1, p2, p3, max;

printf("Enter length and breadth of rectangle 1:");
scanf("%d%d", &l1, &b1);
```

```
printf("Enter length and breadth of rectangle 2:");
scanf("%d %d", &l2, &b2);

printf("Enter length and breadth of rectangle 3:");
scanf("%d, %d", &l3, &b3);

printf("Enter length and breadth of rectangle

p1 = 2*(l1 + b1);
p2 = 2* (l2 + b2);
p3 = 2* (l3 + b3);

printf("\nPerimeter of rectangle 1 = %d", p1);
printf("\nPerimeter of rectangle 2= %d", p2);
printf("\nPerimeter of rectangle 3= %d", p3);

max = (p1 > p2) ? ((p1 > p3) ? p1 : p3) : ((p2 > p3) ?
  p2 : p3);

printf("\nThe highest perimeter is : %d\n", max);

return0;
}
```



```c
#include <stdio.h>

int main() {
  int year, i, total_days = 0;
  int leap_years, normal_years;
  int day;

  printf("Enter the year: ");
  scanf("%d", &year);

  // 01/01/01 is Monday (day = 1)
  // We will count days from year 1 to (year - 1)
  for (i = 1; i < year; i++) {
    if ((i % 400 == 0) || (i % 4 == 0 && i % 100 != 0))
      total_days += 366; // Leap year
    else
      total_days += 365; // Normal year
  }

  // Find day of the week (Monday = 1, Tuesday = 2, ... Sunday = 0)
  day = (total_days % 7 + 1) % 7;

  printf("On 1st January %d, the day is: ", year);
  switch (day) {
    case 1: printf("Monday\n"); break;
    case 2: printf("Tuesday\n"); break;
    case 3: printf("Wednesday\n"); break;
    case 4: printf("Thursday\n"); break;
```

```
PS C:\Users\ACER\Desktop\itscbe> gcc 3.4.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
Enter the year: 2000
On 1st January 2000, the day is: Saturday
PS C:\Users\ACER\Desktop\itscbe>
```

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```
}
return0;
}
```

4. According to the gregorian calendar. It was Monday on the date 01/01/01. If Any year is input through the keyboard write a program to find out what is the day on. 1st January of this year.

```
# include <stdio.h>

int main() {
    int year, i, day = 1;
    printf("Enter year :");
    scanf("%d", &year);

for(i=1;i<year; i++) {
    if ((i % 400 ==0) || (i % 4==0 && i % 100!=0)) {
        day = (day + 2) % 7;
    } else {
        day = (day+1) % 7;
    }
}
printf("01/01/%d is", year);
switch (day) {
```

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```c
#include <stdio.h>

int main() {
    float x1, y1, x2, y2, x3, y3, area;

    // Input coordinates
    printf("Enter coordinates of first point (x1 y1): ");
    scanf("%f %f", &x1, &y1);

    printf("Enter coordinates of second point (x2 y2): ");
    scanf("%f %f", &x2, &y2);

    printf("Enter coordinates of third point (x3 y3): ");
    scanf("%f %f", &x3, &y3);

    // Formula to check collinearity using area of triangle
    // Area = 0.5 * |x1(y2 - y3) + x2(y3 - y1) + x3(y1 - y2)|
    area = x1*(y2 - y3) + x2*(y3 - y1) + x3*(y1 - y2);

    if (area == 0)
        printf("The points are Collinear.\n");
    else
        printf("The points are NOT Collinear.\n");

    return 0;
}
```

```
PS C:\Users\ACER\Desktop\itscbe> gcc 3.3.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
Enter coordinates of first point (x1 y1): 4
5
Enter coordinates of second point (x2 y2): 4
3
Enter coordinates of third point (x3 y3): 3
5
The points are NOT Collinear.
PS C:\Users\ACER\Desktop\itscbe>
```

```
else {
    printf("Category : Undefined Range\n");
  }
  return 0;
}
```

3. Wap to check if three points (x_1, y_1) (x_2, y_2) and $(x_3,$
are collinear or not.

```
#include<stdio.h>

int main() {
  int x1, y1, x2, y2, x3, y3;

printf("Enter x1 y1 : ");
scanf("%d %d", &x1, &y1);
printf("Enter x2 y2 : ");
scanf("%d %d, &x2, &y2);
printf("Enter x3 y3 : ");
scanf(" %d %d, &x3, &y3);

if ((x2-x1) * (y3-y1) == (y2-y1) * (x3-x1)) {
   printf("The points are collinear.\n");
} else {
   printf("The points are NOT Collinear.\n");
```

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```c
#include <stdio.h>

int main() {
  float weight, height, bmi;

  // Input weight and height
  printf("Enter your weight (in kilograms): ");
  scanf("%f", &weight);

  printf("Enter your height (in meters): ");
  scanf("%f", &height);

  // Calculate BMI
  bmi = weight / (height * height);

  // Display BMI value
  printf("\nYour BMI is: %.2f\n", bmi);

  // Determine category
  if (bmi < 18.5)
    printf("You are Underweight.\n");
  else if (bmi >= 18.5 && bmi < 25)
    printf("You are Normal weight.\n");
  else if (bmi >= 25 && bmi < 30)
    printf("You are Overweight.\n");
  else
    printf("You are Obese.\n");
```

```
PS C:\Users\ACER\Desktop\itscbe> gcc 3.2.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
Enter your weight (in kilograms): 49
Enter your height (in meters): 150

Your BMI is: 0.00
You are Underweight.
PS C:\Users\ACER\Desktop\itscbe>
```